Domain-Adversarial Training Under Label Shift Can Silently Invert Out-of-Distribution Detection

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Abstract

Domain-adversarial training (DANN) is a standard remedy for site and device shortcuts in medical imaging, and deployed models increasingly rely on out-of-distribution (OOD) monitors to flag unfamiliar inputs. We show that the two can conflict silently. On skin-lesion classification with a lesion-disjoint ISIC 2019 split, gradient reversal toward smartphone images drives Mahalanobis OOD detection from AUROC 0.825 to 0.381 (95% CI $[0.358, 0.404]$), and below chance on three clinical datasets no part of training saw (Fitzpatrick17k 0.405; MED-NODE and SD-198 [PENDING: last-checkpoint values]), while in-distribution balanced accuracy stays at 0.47–0.48 [PENDING: paired CI, calibration]. The inversion is caused by label shift between the domains the adversary aligns: changing only the class mix the adversary sees moves unseen-domain AUROC by 0.24–0.37, and the effect persists at matched domain leakage. Mechanistically, DANN pulls unfamiliar images toward the majority class; under label shift it reaches chance-level unweighted domain leakage by misaligning classes, while the domain stays decodable within each class. The consequence is clinical: the share of melanomas from unseen sources predicted benign nevus with confidence above 0.9 rises from 8% to 49–81%. Importance-weighted alignment helps in proportion to how well the target class priors are known: with weights estimated online from unlabelled target data it recovers 32–45% of the lost AUROC at matched leakage, and with priors from 50 labelled target images detection returns to its pre-adaptation level, at a cost of 0.08 in-distribution balanced accuracy. OOD monitoring should be re-validated after domain adaptation, and domain leakage should be measured within class.

Keywords: domain-adversarial training; out-of-distribution detection; label shift; calibration; skin lesion classification; deployment monitoring

1. Introduction

Skin lesion classifiers learn acquisition artefacts as readily as pathology: ink markings, rulers, dermatoscope type and contributing site all correlate with diagnosis in public datasets (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often by domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). A deployed model also needs a second safeguard: an out-of-distribution (OOD) monitor that flags inputs unlike its training data so that a clinician reviews them. This paper asks what invariance training does to that monitor.

In our setting it can turn the monitor upside down. As the adversarial weight grows, OOD detection does not merely weaken; it falls below chance, so that images from clinics the model has never seen score as *more* familiar than its own test data. Nothing in a standard evaluation reveals this: in-distribution accuracy stays flat [PENDING: calibration

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at the last checkpoint], and the domain-leakage probe, the quantity the method is meant to reduce, improves. The cost falls on the patients a monitor exists to protect: under the same training, melanomas from unseen clinics are predicted benign nevus with confidence above 0.9 up to ten times more often, and the OOD monitor that should route them to a clinician ranks them as routine.

We trace the failure to label shift. ISIC 2019 is 50.8% nevus, the smartphone images the adversary aligns against are 12.3% nevus, and matching the marginal feature distributions of domains with different class priors forces the representation to misplace one domain's classes (Zhao et al., 2019; Tachet des Combes et al., 2020). We show the consequence for OOD monitoring with an intervention that changes only the class mix the adversary sees, and we locate the mechanism in the feature space.

Contributions.

1. **The failure.** Under a lesion-disjoint split, gradient reversal inverts Mahalanobis OOD detection on held-out smartphone images and on three clinical datasets that no branch of training saw, with flat in-distribution accuracy [PENDING: and calibration], under two training recipes whose accuracy is within 0.06 of a standard ERM baseline on the same split.

2. **The cause.** Setting the class mix seen by the adversary to that of the target removes the inversion on unseen domains (AUROC difference 0.24–0.37); a regression at matched domain leakage attributes the effect to label shift rather than to deeper invariance.

3. **The mechanism and a measurement lesson.** DANN pulls images that differ from the source along the domain direction toward the majority class. Under label shift, an unweighted domain probe reports chance-level leakage while a within-class probe still reads the domain: invariance should be measured within class.

4. **Consequence and mitigation.** Melanomas from unseen sources are called benign nevi with high confidence up to ten times more often. Importance weighting with label-free prior estimates recovers part of the loss; in a post hoc analysis, priors from 50 labelled target images bring detection back to the oracle level, at a cost in in-distribution accuracy.

2. Related work

Shortcuts and invariance in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion masked out (Bissoto et al., 2020), and performance degrades on darker skin (Daneshjou et al., 2022). Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) are the usual remedies; our dual-encoder model is a domain separation network and is a vehicle for the intervention, not a contribution.

Invariance under label shift. When class priors differ between domains, matching marginal feature distributions cannot also keep the joint error small (Zhao et al., 2019);

importance-weighted alignment (IWDAN, IWCDAN) corrects for this with estimated target class priors (Tachet des Combes et al., 2020), which can be estimated without target labels by black-box shift estimation (Lipton et al., 2018) or EM (Saerens et al., 2002). These results concern classification error. We study what the same misalignment does to OOD monitoring, and show that a standard domain-leakage probe misreports invariance under label shift.

OOD detection after adaptation. Full-spectrum OOD detection (Yang et al., 2023; Zhang et al., 2023) asks that covariate-shifted in-distribution inputs be accepted; our smartphone and clinical images fall in that cell, and Section 5 argues that acceptance is safe only when the model also classifies them. Calibration degrades under shift (Ovadia et al., 2019) and after domain adaptation (Wang et al., 2020). Closest to our observation, Wu et al. report Mahalanobis and ODIN AUROC below 0.5 on rotated MNIST after adaptation with an MMD-based method (DAN) (Wu et al., 2023), as a baseline in a study of continually adaptive detection. We show the inversion on medical images, identify label shift as its cause, and characterise the mechanism.

3. Materials and methods

3.1. Data and splits

ISIC 2019 (Tschandl et al., 2018) (25,331 dermoscopic images, eight classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC) is the in-distribution source. We split it 60/20/20 into train, validation and test *grouped by* `lesion_id` (16,211 / 4,053 / 5,067 images), because 60% of test images in the image-level split share a lesion with training (Tran, 2027). **PAD-UFES-20** (Pacheco et al., 2020) (2,298 smartphone photographs, six of the eight classes) is the domain the adversary aligns against. Its labels are never used in any loss. We split it by patient into `pad_adv` (1,582 images, 961 patients), seen by the adversarial and context branches, and `pad_heldout` (716 images, 412 patients), never seen.

Unseen domains. Four clinical-photograph datasets enter no part of training: Fitzpatrick17k (Groh et al., 2021) (3,887 images obtained), MED-NODE (170 images), SD-198 (6,584 images) and SCIN (10,406 images, 5,033 cases) [PENDING: cite MED-NODE, SD-198, SCIN; licences: SCIN CC BY 4.0, MED-NODE none stated, SD-198 official source]. SCIN is mostly inflammatory conditions; only 0.6% of its images map to the ISIC classes, so we use it only as an OOD set for detection and report it separately; no label-based analysis is done on SCIN. Images are cropped to a lesion-centred box and resized to 224×224.

3.2. Model and training

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, z_ℓ^{norm}) used for diagnosis through a linear head, and a context latent $z_c \in \mathbb{R}^{64}$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the

latents. AdamW (learning rate 10^{-4}; adversary $\times 30$, classifier $\times 0.2$), 40 epochs, domain-balanced batches. We sweep $\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ with everything else fixed (seeds 42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 elsewhere). A second recipe (A4: $z_\ell \in \mathbb{R}^{64}$, classifier $\times 1$, class-weighted loss) serves as a robustness check.

Checkpoints. We report the last checkpoint. Validation balanced accuracy (eight classes, chance 0.125) has a lesion-clustered bootstrap SE of 0.023–0.026, and the best-validation and last checkpoints differ by less than one SE, so best-validation selection picks early epochs by noise. Best-validation values are given for comparison.

Competence. In-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ is 0.496 (main recipe) and 0.520 (A4), 0.04–0.06 below a standard single-encoder ERM on the same split (0.556 [PENDING: ERM reference at 5 seeds, with SE]) and four times chance (0.125). Lesion-disjoint eight-class ISIC 2019 is hard; the image-level split inflates the same model's balanced accuracy by 0.20 (Tran, 2027).

3.3. Interventions on label shift

Class mix seen by the adversary. Keeping the classification loss on all of ISIC, we resample only the ISIC images fed to the adversarial branch so that their class mix is ISIC's own (100%, the default), halfway to PAD's (50%), or PAD's (0%, no label shift between the aligned domains), at $\lambda_{\text{adv}} \in \{0.5, 2\}$ with five seeds (three at 50%).

Importance-weighted alignment. IWDAN and IWCDAN (Tachet des Combes et al., 2020) reweight the source in the adversarial loss by target class priors. In the original method the weights are estimated online during training from unlabelled target predictions; as alternatives we use priors from 50 labelled `pad_adv` images (a few-label setting) and the true `pad_adv` priors (oracle, not deployable). Black-box shift estimates (Lipton et al., 2018) on the final checkpoints serve only as a diagnostic, and an EM estimator (Saerens et al., 2002) failed as implemented and is not used. CDAN (Long et al., 2018) is a class-conditional adversarial baseline.

3.4. Measurements

OOD detection. Class-conditional Mahalanobis with shared covariance (Lee et al., 2018) (primary), k-NN (Sun et al., 2022), maximum cosine to a class centroid, maximum softmax probability (Hendrycks and Gimpel, 2017) and energy (Liu et al., 2020), all fit on ISIC train. AUROC treats OOD as the positive class, so 0.5 is chance and values below 0.5 mean OOD images score as more familiar than ID test images. The five scores are functions of the same latent, so they are not independent evidence.

Inversion. We call a cell inverted when the cluster-bootstrap 95% CI of the seed-mean Mahalanobis AUROC lies below 0.5 and at least three seeds' own intervals do (all three in cells with three seeds). Bootstrap draws (2,000) resample ISIC test by lesion and OOD sets by patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k) or case (SCIN); MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow. Runs with in-distribution balanced accuracy below 0.25 are excluded as degraded and reported.

Table 1: The adversarial-weight sweep (main recipe, lesion-disjoint ISIC split, last checkpoint). Mean ± s.d. over five seeds per λ_{adv} (42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 otherwise). Leakage: linear domain probe on held-out images (chance 0.5). ID: balanced accuracy, eight classes (chance 0.125). Mahalanobis AUROC, OOD positive; k: seeds whose interval lies below 0.5. Best-val: the best-validation checkpoint, for comparison.

λ_{adv}	leakage	ID bal. acc	`pad_heldout`	k	Fitzpatrick17k	k	`pad_heldout` (best-val)
0	0.842 ± 0.050	0.481 ± 0.016	0.825 ± 0.022	0/5	0.712 ± 0.013	0/5	0.808 ± 0.027
0.25	0.502 ± 0.002	0.481 ± 0.010	$\mathbf{0.445 \pm 0.022}$	4/5	$\mathbf{0.396 \pm 0.032}$	5/5	0.525 ± 0.121
0.5	0.508 ± 0.010	0.484 ± 0.005	$\mathbf{0.415 \pm 0.016}$	5/5	$\mathbf{0.377 \pm 0.046}$	5/5	0.426 ± 0.008
1	0.505 ± 0.005	0.482 ± 0.017	$\mathbf{0.390 \pm 0.022}$	5/5	$\mathbf{0.388 \pm 0.039}$	5/5	0.387 ± 0.016
2	0.507 ± 0.009	0.472 ± 0.018	$\mathbf{0.381 \pm 0.023}$	5/5	$\mathbf{0.405 \pm 0.031}$	5/5	0.403 ± 0.028
4	0.527 ± 0.035	0.480 ± 0.016	$\mathbf{0.416 \pm 0.034}$	5/5	$\mathbf{0.354 \pm 0.066}$	5/5	0.433 ± 0.020
8	0.520 ± 0.028	0.480 ± 0.018	$\mathbf{0.439 \pm 0.020}$	5/5	$\mathbf{0.383 \pm 0.036}$	5/5	0.454 ± 0.044

Domain leakage. Balanced accuracy (chance 0.5) of a logistic probe for the source dataset on frozen z_ℓ^{norm}. The probe used in earlier work scores ISIC test against all of PAD, including images the adversary trained on. We therefore also report probes on held-out images only (linear, one-hidden-layer MLP, k-NN), a class-balanced probe that reweights both domains to a common class mix, and a within-class probe averaged over classes.

Calibration and diagnosis. Expected calibration error (15 bins) on ISIC test and on `pad_heldout` (six shared classes), and zero-shot balanced accuracy on `pad_heldout`. Out-of-domain ECE is close to mean confidence minus accuracy here, so these are views of one phenomenon.

4. Results

4.1. OOD detection inverts while in-distribution metrics stay flat

The sweep. At the last checkpoint, Mahalanobis AUROC on `pad_heldout` falls from 0.825 at $\lambda_{\text{adv}} = 0$ to 0.445 at $\lambda_{\text{adv}} = 0.25$ and stays between 0.381 and 0.445 at every larger weight, meeting the inversion criterion at all six non-zero weights; on Fitzpatrick17k it falls from 0.712 to 0.354–0.405 (Table 1). At $\lambda_{\text{adv}} = 2$ the seed-mean interval is $[0.358, 0.404]$ on `pad_heldout` and $[0.392, 0.419]$ on Fitzpatrick17k. In-distribution balanced accuracy stays within 0.472–0.484 [PENDING: paired $\lambda_{\text{adv}} = 2 - \lambda_{\text{adv}} = 0$ CI and ID/OOD ECE at the last checkpoint]. All five scores fall below 0.5 on `pad_heldout` at $\lambda_{\text{adv}} = 2$ (k-NN 0.427, cosine 0.420, MSP 0.455, energy 0.468; best-validation checkpoint); the standard k-th-neighbour k-NN gives the same values within 0.01. The best-validation checkpoint gives the same picture from $\lambda_{\text{adv}} = 0.5$ on, but at $\lambda_{\text{adv}} = 0.25$ it selects an earlier epoch and the inversion is not yet visible (0.525, seed s.d. 0.121; an observation made after the sweep).

Unseen domains. The inversion is not confined to the domain the adversary was trained against (Table 2). On three clinical datasets that entered no part of training, Mahalanobis AUROC falls from 0.717 to 0.402 (Fitzpatrick17k), 0.536 to 0.306 (MED-NODE) and 0.713 to 0.375 (SD-198) at $\lambda_{\text{adv}} = 2$, each meeting the inversion criterion. On SCIN, whose images

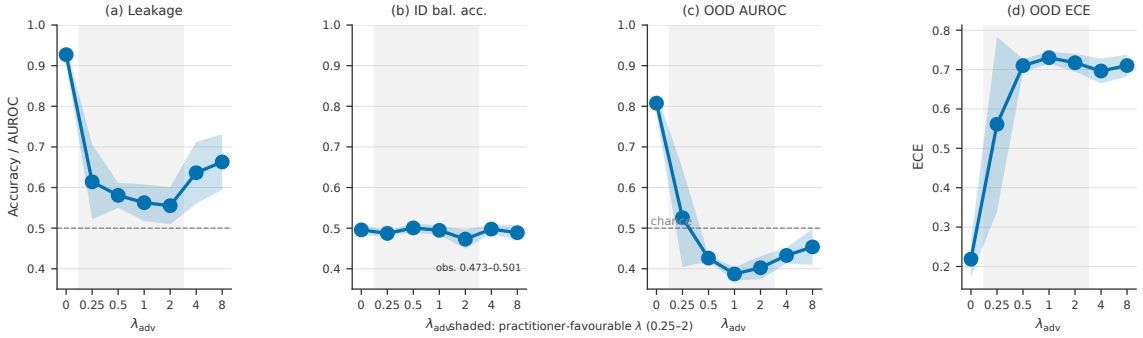


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

Table 2: Unseen domains: Mahalanobis AUROC at $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$, mean ± s.d. over five seeds, with the number of seeds whose interval lies below 0.5. Main recipe, best-validation checkpoint [PENDING: MED-NODE/SD-198/SCIN at the last checkpoint]; A4, last checkpoint. MED-NODE and SD-198 are resampled by image. –: not bootstrapped.

recipe	set	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$	seeds < 0.5	inverted
main	`pad_heldout`	0.808 ± 0.027	0.403 ± 0.028	5/5	yes
main	Fitzpatrick17k	0.717 ± 0.034	0.402 ± 0.018	5/5	yes
main	MED-NODE	0.536 ± 0.037	0.306 ± 0.071	5/5	yes
main	SD-198	0.713 ± 0.037	0.375 ± 0.026	5/5	yes [PENDING: SD-198 cluster CI]
main	SCIN	0.726 ± 0.021	0.559 ± 0.034	0/5	no
A4	`pad_heldout`	0.786 ± 0.019	0.452 ± 0.014	[PENDING: k]	yes
A4	Fitzpatrick17k	0.821 ± 0.023	0.458 ± 0.061	[PENDING: k]	yes

are mostly inflammatory conditions outside the ISIC label space, detection falls from 0.726 to 0.559 without inverting.

A second recipe. With a 64-dimensional latent, unweighted cross-entropy replaced by a class-weighted loss and the last checkpoint (A4), detection again inverts at $\lambda_{\text{adv}} = 0.5$ and $\lambda_{\text{adv}} = 2$ on `pad_heldout` (0.468 $[0.440, 0.495]$; 0.452 $[0.425, 0.478]$) and Fitzpatrick17k (0.444; 0.458), with in-distribution balanced accuracy unchanged (0.520, 0.520, 0.510). On `pad_heldout` all five scores' intervals lie below 0.5; on Fitzpatrick17k k-NN (0.521) and energy (0.516) only collapse. A plain single-encoder DANN collapses to chance (0.51–0.56) without inverting (Appendix A).

4.2. Label shift between the aligned domains causes the inversion

Intervention. Changing only the class mix of the ISIC images the adversary sees switches the inversion on unseen domains on and off (Table 3). At $\lambda_{\text{adv}} = 2$, with the adversary's ISIC images resampled to PAD's class mix (shift 0%), Mahalanobis AUROC is 0.691 on Fitzpatrick17k, 0.541 on MED-NODE and 0.694 on SD-198, none inverted; with ISIC's own mix (shift 100%) it is 0.340, 0.298 and 0.325, inverted in every seed. Paired by seed, removing the label shift raises AUROC by 0.361, 0.243 and 0.372. The intermediate level falls between the two (Fitzpatrick17k 0.482 at $\lambda_{\text{adv}} = 0.5$, 0.441 at $\lambda_{\text{adv}} = 2$). On `pad_heldout`, the adversary's own target domain, a mild inversion remains without label shift (0.458; paired difference +0.093). One of five shift-0% runs collapsed (in-distribution balanced accuracy 0.139) and is excluded.

Matched leakage. Removing label shift also leaves more domain information in the latent (leakage 0.791 vs 0.634), so the intervention alone does not separate label shift from depth of invariance. A regression of unseen-domain AUROC on label-shift level and leakage, with recipe as a covariate (variance inflation ≈ 1.04), gives label-shift coefficients of -0.166 $[-0.217, -0.115]$ (Fitzpatrick17k), -0.107 $[-0.148, -0.065]$ (MED-NODE) and -0.172 $[-0.221, -0.123]$ (SD-198) for a change from no shift to full shift at fixed leakage [PENDING: confirm shift coding and resampling unit]. On `pad_heldout` the coefficient is $+0.041$ $[-0.033, +0.110]$. Comparing runs at equal leakage across recipes (A4 at shift 100%, leakage 0.78, against the main recipe at shift 0%, leakage 0.79) gives the same direction (+0.23 to +0.26).

A natural example without DANN. Label shift alone can invert a class-conditional Mahalanobis score. Training on BCN and treating HAM as OOD, Mahalanobis AUROC is already 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, while k-NN gives 0.658. HAM-heldout is 67% nevus against 31% in BCN test; reweighting HAM to BCN's class mix gives 0.489 $[0.453, 0.524]$, and no class-matched AUROC differs from 0.5 (post hoc). In the main setting the OOD sets are *less* nevus-heavy than ISIC, so detection works at $\lambda_{\text{adv}} = 0$; DANN creates the condition by moving them toward nevus.

4.3. Mechanism: misaligned classes behind chance-level leakage

Unfamiliar images move to the majority class. At $\lambda_{\text{adv}} = 2$ with full label shift, the share of `pad_heldout` images whose nearest class centroid is nevus rises from 0.28 to 0.63, and to 0.75–0.85 on the unseen sets; without label shift it stays at 0.12 and 0.39–0.49. Images of non-nevus classes are pulled into the nevus region: split by true class, non-nevus OOD images fall to AUROC 0.42 (main recipe) and 0.47 (A4), whereas images still nearest a non-nevus centroid stay at 0.62 and 0.64. Because pooled AUROC is the class-share-weighted mean of per-class AUROCs, and nevus is half the ISIC test set, this displacement is enough to push the pooled score below chance.

The displacement follows the domain direction. For each OOD image we measured how far it lies along the direction separating ISIC from PAD in the $\lambda_{\text{adv}} = 0$ model, and how much closer to its nearest ID centroid it moves at $\lambda_{\text{adv}} = 2$. The two correlate on every lesion dataset (Spearman 0.33–0.47), while the deviation orthogonal to that direction does not predict the pull ($|\rho| \leq 0.07$ on four of five sets). Between two $\lambda_{\text{adv}} = 0$ models of

Table 3: Label-shift intervention: Mahalanobis AUROC (main recipe, last checkpoint) by the class mix the adversary sees. Shift 100%: ISIC's own mix; 0%: PAD's mix. Leakage: linear probe on ISIC test against all of PAD, as in earlier work. Shift 0% at $\lambda_{\text{adv}} = 2$: four seeds after excluding one degraded run (in-distribution balanced accuracy $0.139 < 0.25$); shift 50%: three seeds.

λ_{adv}	shift	leakage	`pad_heldout`	Fitzpatrick17k	MED-NODE	SD-198
0	–	0.928	0.825	0.712	[PENDING:]	[PENDING:]
0.5	0%	0.790	0.499	0.674	[PENDING:]	[PENDING:]
0.5	50%	0.629	0.490	0.482	[PENDING:]	[PENDING:]
0.5	100%	0.628	0.361	0.385	[PENDING:]	[PENDING:]
2	0%	0.791	0.458	0.691	0.541	0.694
2	50%	0.658	0.454	0.441	[PENDING:]	[PENDING:]
2	100%	0.634	0.368	0.340	0.298	0.325

different seeds the same correlation is smaller (0.11–0.29); the DANN-specific excess exceeds 0.15 on all four unseen datasets (0.18–0.33) but not on `pad_heldout` (0.057), and is larger in the pre-projection backbone (0.24–0.45 on all five).

Chance-level leakage hides the misalignment. The domain probe of Table 1 is scored partly on images the adversary trained on. On held-out images a linear probe reads the domain at chance from $\lambda_{\text{adv}} = 0.5$ on (0.502–0.506), but a k-NN probe still reads it (0.55–0.61), and under label shift an unweighted probe misreports invariance (Table 4). With full label shift DANN reaches 0.510 on an unweighted held-out probe, yet within each class the domain remains decodable (0.618), more so than in any other condition. Correcting the class mix the adversary aligns (shift 0%, or importance weighting with the true priors) raises the unweighted reading (0.63–0.64) while lowering the within-class one (0.56–0.57). Under label shift, DANN matches the marginal distributions by misaligning classes, as the theory of Zhao et al. (2019) implies; leakage should therefore be measured within class (post hoc).

4.4. Clinical consequence and mitigation

Melanomas called benign. With a threshold accepting 95% of ISIC validation images, almost all unseen-domain malignant images are accepted at every λ_{adv}, so the monitor does not catch them in either case. What changes is the classifier: the share of melanomas predicted nevus with confidence above 0.9 rises from 7.5% to 48.7% on Fitzpatrick17k and from 8.1% to 81.2% on SD-198 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ [PENDING: n and CIs per cell; label map].

No fix at scoring time. Reversing the sign of the score is not a remedy: at $\lambda_{\text{adv}} \geq 0.5$ it reaches only 0.55–0.63 on `pad_heldout` and Fitzpatrick17k with a false-positive rate of 0.84–0.95 at 95% true-positive rate [PENDING: verify FPR range in s_flip], and at $\lambda_{\text{adv}} = 0$ it would turn 0.808 into 0.192. Relative Mahalanobis (Ren et al., 2021) and per-class standardised distances stay below 0.5 on all three unseen datasets at $\lambda_{\text{adv}} = 2$.

Table 4: Domain leakage on held-out images under three probes, and unseen-domain detection ($\lambda_{\text{adv}} = 2$, main recipe, last checkpoint). Unweighted: as usually reported; class-balanced: both domains reweighted to a common class mix; within-class: class-size-weighted mean of per-class probes. Fitz: Mahalanobis AUROC on Fitzpatrick17k.

condition	unweighted	class-balanced	within-class	Fitz
DANN, shift 100%	0.510	0.602	0.618	0.340
DANN, shift 0%	0.629	0.539	0.560	0.691
IWDAN (online)	0.534	0.562	0.597	0.508
IWDAN (50 labels)	0.653	0.545	0.583	0.741
IWDAN (true priors)	0.645	0.549	0.576	0.717

Importance-weighted alignment. IWDAN reweights the source in the adversarial loss by estimated target class priors, and how well it works depends on the estimate. With weights updated online from unlabelled target predictions, as in the original method, the implied target priors end 0.43–0.55 (total variation; mean 0.515) from the truth: nevus is down-weighted from 0.51 toward 0.23–0.28, against a true 0.10. IWDAN then keeps leakage at the DANN level (0.642 vs 0.634) and recovers 32% of the lost AUROC on `pad_heldout` (0.513) and 45% on Fitzpatrick17k (0.508), while MED-NODE (0.350) and SD-198 (0.488) stay at or below chance. *EM* estimation was unstable: in two of five seeds it placed mass on classes absent from PAD, so we treat it as not having run correctly. With priors from 50 *labelled target images* (total variation 0.10–0.14), detection on the unseen domains returns to the level obtained with the true priors (Fitzpatrick17k 0.741 vs 0.717, SD-198 0.734 vs 0.693, MED-NODE 0.606 vs 0.563; `pad_heldout` 0.619 vs 0.647), and within-class leakage is lower than under DANN (0.58 vs 0.62). This is a promising but post hoc mitigation: neither the 50-label nor the true-prior variant meets our leakage criterion, because unweighted leakage rises (0.81 and 0.78 against 0.63), and both cost in-distribution balanced accuracy (0.39–0.40, against 0.48 both under DANN and before adaptation). Read descriptively, detection improves as the prior error falls (0.52: partial; 0.10–0.14: oracle level; 0: oracle), but our planned rank-correlation test across estimators was not supported (Spearman -0.33, $[-0.80, +0.28]$), with EM contributing invalid priors. As a diagnostic, black-box shift estimates computed on the final checkpoints were further from the truth (0.65–0.76): their assumption of unchanged class-conditional predictions fails across these domains, with target BCC and AK images predicted as MEL or NV. Label-free prior estimates thus remain about 0.5 total variation off and recover detection only in part; 50 labels bring the error to 0.10–0.14 and detection to the oracle level [PENDING: IWCDAN, CDAN, $\lambda_{\text{adv}} = 0.5$ cells; per-method details to appendix].

5. Discussion

Tolerance without generalisation. Full-spectrum OOD detection asks that covariate-shifted inputs be accepted (Yang et al., 2023). After DANN our models accept unseen-clinic images more readily than their own test data while classifying them at a zero-shot balanced

accuracy of 0.254 at $\lambda_{\text{adv}} = 2$ (chance 0.167; best-validation checkpoint). A tolerant detector would score such inputs near 0.5; scoring them below 0.5 is not tolerance but a monitor that has lost its meaning. Claims of tolerance to covariate shift should report cross-domain accuracy on the same inputs.

Why label shift. Matching marginal feature distributions across domains with different class priors cannot keep both domains' classes in place (Zhao et al., 2019). Our results show what this misplacement does to a distance-based monitor: unfamiliar images that differ along the domain direction land in the majority class's region, where a class-conditional score reads them as typical. The BCN→HAM case shows that a nevus-heavy OOD set alone is enough to invert such a score; DANN produces that condition in feature space when the aligned domains differ in class mix. Without label shift the inversion stays on the adversary's own target domain, where a smaller within-class pull remains. The same misplacement explains why an unweighted domain probe reports chance-level leakage: it measures marginal, not class-conditional, alignment.

Checkpoint selection can hide the failure. As an observation made after the sweep, at $\lambda_{\text{adv}} = 0.25$ the last checkpoint is inverted on both `pad_heldout` and Fitzpatrick17k, whereas best-validation selection picks earlier epochs at which the seeds still disagree (seed s.d. 0.121). Validation accuracy cannot tell these checkpoints apart, so a model selected in the usual way can look safe because it was stopped before the failure formed.

No fix at scoring time. Reversing the score works only if one knows in advance that the model is inverted; at $\lambda_{\text{adv}} = 0$ it would turn a working monitor into a broken one, and at $\lambda_{\text{adv}} \geq 0.5$ it leaves most OOD images accepted at a 95% ID-acceptance threshold. Scores designed to be less sensitive to class structure, such as Relative Mahalanobis, also stay below chance. The remedy has to act on training, through the class priors the adversary aligns.

Recommendations. After any domain-alignment step: (i) re-validate OOD detection on images from unseen sources, per class as well as pooled; (ii) report domain leakage within class, not only unweighted; (iii) compare class priors between the aligned domains, and if they differ, correct for them, which in our setting needed a small labelled sample from the target domain; and (iv) report results at more than one checkpoint.

6. Limitations

Scope. The inversion is shown in dermatology only. We attempted two other modalities and neither experiment took place: on Camelyon17 no adversarial setting we tried removed hospital information while preserving the classifier, and on chest radiographs the detector was near chance on the OOD set before any adaptation (AUROC 0.522), so there was nothing to invert. **Architecture.** The main results use a dual-encoder model; the A4 recipe changes the latent size and loss, but a plain single-encoder DANN collapses detection to chance without inverting, and whether deeper invariance would invert it is untested [PENDING: R1-hi / P9]. **Competence.** In-distribution balanced accuracy is 0.04–0.06 below a standard ERM on the same lesion-disjoint split [PENDING: ERM at 5 seeds]. **Statistics.** MED-NODE and SD-198 carry no patient identifiers and are resampled by

image, so their intervals may be too narrow; SCIN is used only as an OOD set because almost none of it maps to the ISIC classes. The five OOD scores are functions of one latent and are not independent evidence. **Mechanism.** A two-domain Gaussian toy model with label shift did not reproduce the inversion, so the conditions we identify may not be sufficient in general. **Warning signals.** No statistic computed on in-distribution data alone flagged the inverted runs; the share of deployment images predicted as the majority class tracked inversion (Spearman -0.73 [PENDING: update to the n=64 corpus]) but needs unlabelled target data. **Mitigation.** The few-label remedy is a post hoc analysis, costs in-distribution accuracy, and did not meet our leakage criterion. **Data.** We obtained 3,887 of 16,577 Fitzpatrick17k images.

7. Conclusion

Domain-adversarial training can make an OOD monitor rank images from unseen clinics as more familiar than its own test data, while every routinely reported in-distribution metric stays flat. In our setting the cause is label shift between the domains the adversary aligns: it misplaces unfamiliar images into the majority class and hides the misalignment from the usual leakage probe. Checking OOD detection on unseen sources after adaptation, measuring leakage within class and correcting for class priors are inexpensive steps that make the failure visible and, with a small labelled target sample, reduce it.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

References

Alceu Bissoto, Eduardo Valle, and Sandra Avila. Debiasing skin lesion datasets and models? Not so fast. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020.

Konstantinos Bousmalis, George Trigeorgis, Nathan Silberman, Dilip Krishnan, and Dumitru Erhan. Domain separation networks. In *Advances in Neural Information Processing Systems*, 2016.

Roxana Daneshjou, Kailas Vodrahalli, Roberto A. Novoa, Melissa Jenkins, Weixin Liang, Veronica Rotemberg, Justin Ko, Susan M. Swetter, Elizabeth E. Bailey, Olivier Gevaert, Pritam Mukherjee, Michelle Phung, Kiana Yekrang, Bradley Fong, Rachna Sahasrabudhe, Johan A. C. Allerup, Utako Okata-Karigane, James Zou, and Albert S. Chiou. Disparities in dermatology AI performance on a diverse, curated clinical image set. *Science Advances*, 8(32), 2022.

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.

Matthew Groh, Caleb Harris, Luis Soenksen, Felix Lau, Rachel Han, Aerin Kim, Arash Koochek, and Omar Badri. Evaluating deep neural networks trained on clinical images in dermatology with the Fitzpatrick 17k dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2021.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems*, 2018.

Zachary C. Lipton, Yu-Xiang Wang, and Alexander J. Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.

Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems*, 2020.

Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems*, 2018.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019.

Andre G. C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, Fábio C. R. Alves Jr, José G. M. Esgario, Alana C. Simora, Pedro B. C. Castro, Felipe B. Rodrigues, Patricia H. L. Frasson, Renato A. Krohling, Helder Knidel, Maria C. S. Santos, Rachel B. do Espírito Santo, Telma L. S. G. Macedo, Tania R. P. Canuto, and Luiz F. S. de Barros. PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32, 2020.

Jie Ren, Stanislav Fort, Jeremiah Liu, Abhijit Guha Roy, Shreyas Padhy, and Balaji Lakshminarayanan. A simple fix to Mahalanobis distance for improving near-OOD detection. *arXiv preprint arXiv:2106.09022*, 2021.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1): 21–41, 2002.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning*, 2022.

Remi Tachet des Combes, Han Zhao, Yu-Xiang Wang, and Geoffrey Gordon. Domain adaptation with conditional distribution matching and generalized label shift. In *Advances in Neural Information Processing Systems*, 2020.

Duc Vinh Tran. Your OOD monitor has seen these patients: Group overlap miscalibrates deployment monitors in medical imaging, 2027. Under review at MIDL 2027. Co-author list not final.

Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5, 2018. See also Combalia et al., BCN20000, arXiv:1908.02288, 2019, and the ISIC 2019 challenge.

Ximei Wang, Mingsheng Long, Jianmin Wang, and Michael I. Jordan. Transferable calibration with lower bias and variance in domain adaptation. In *Advances in Neural Information Processing Systems*, 2020.

Julia K. Winkler, Christine Fink, Ferdinand Toberer, Alexander Enk, Teresa Deinlein, Rainer Hofmann-Wellenhof, Luc Thomas, Aimilios Lallas, Andreas Blum, Wilhelm Stolz, and Holger A. Haenssle. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. *JAMA Dermatology*, 155(10):1135–1141, 2019.

Xinheng Wu, Jie Lu, Zhen Fang, and Guangquan Zhang. Meta OOD learning for continuously adaptive OOD detection. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. doi: 10.1109/ICCV51070.2023.01773.

Jingkang Yang, Kaiyang Zhou, and Ziwei Liu. Full-spectrum out-of-distribution detection. *International Journal of Computer Vision*, 2023.

Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Kaiyang Zhou, Wayne Zhang, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*, 2023.

Han Zhao, Remi Tachet des Combes, Kun Zhang, and Geoffrey Gordon. On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *PMLR*, pages 7523–7532, 2019.

Appendix A. Other alignment objectives and the single-encoder model

All values are Mahalanobis, k-NN, cosine, MSP and energy AUROC on `pad_heldout` (image-level ISIC split, best-validation checkpoint, patient-clustered intervals), so they are comparable with each other but not with the main tables.

MMD at weight 10,000 reaches leakage 0.836 only by reducing in-distribution balanced accuracy to 0.337, so it never reaches the invariance gradient reversal attains with the classifier intact [PENDING: P11: MMD/MK-MMD at matched invariance]. IRM's low MSP and energy values reflect a classifier that has collapsed (balanced accuracy 0.167 at weight 100), not an inversion of the feature scores.

Table 5: Single-encoder DANN (EfficientNet-B3, one linear head and a domain head; five seeds at $\lambda_{\text{adv}} \in \{0, 2\}$, three elsewhere). Detection collapses to about chance from $\lambda_{\text{adv}} = 0.25$ and does not invert.

λ_{adv}	Mahalanobis	k-NN	cosine	MSP	energy
0	0.700 [0.683, 0.717]	0.935 [0.924, 0.944]	0.875 [0.861, 0.887]	0.857 [0.844, 0.868]	0.901 [0.891, 0.910]
0.25	0.539 [0.514, 0.564]	0.548 [0.526, 0.571]	0.530 [0.508, 0.553]	0.548 [0.529, 0.567]	0.536 [0.515, 0.555]
0.5	0.525 [0.499, 0.549]	0.539 [0.516, 0.561]	0.519 [0.498, 0.542]	0.543 [0.522, 0.563]	0.543 [0.524, 0.561]
1	0.555 [0.530, 0.581]	0.541 [0.519, 0.562]	0.517 [0.495, 0.539]	0.517 [0.497, 0.537]	0.498 [0.479, 0.516]
2	0.521 [0.494, 0.546]	0.542 [0.520, 0.562]	0.514 [0.491, 0.535]	0.540 [0.522, 0.559]	0.529 [0.511, 0.547]
4	0.512 [0.486, 0.538]	0.514 [0.489, 0.539]	0.490 [0.467, 0.512]	0.519 [0.498, 0.540]	0.501 [0.481, 0.521]
8	0.525 [0.502, 0.549]	0.558 [0.536, 0.580]	0.526 [0.505, 0.545]	0.566 [0.545, 0.585]	0.551 [0.531, 0.570]

Table 6: Non-adversarial objectives on the single-encoder model, by weight (three seeds). None inverts detection; MMD lowers it while keeping every score above 0.6.

objective	weight	Mahalanobis	k-NN	cosine	MSP	energy
ERM	–	0.834 [0.819, 0.849]	0.940 [0.929, 0.949]	0.888 [0.874, 0.900]	0.826 [0.813, 0.838]	0.823 [0.809, 0.836]
CORAL	1	0.910 [0.899, 0.920]	0.970 [0.962, 0.977]	0.959 [0.949, 0.967]	0.827 [0.814, 0.839]	0.834 [0.821, 0.847]
CORAL	100	0.998 [0.996, 1.000]	0.998 [0.996, 1.000]	0.999 [0.996, 1.000]	0.944 [0.937, 0.951]	0.629 [0.614, 0.644]
MMD	1	0.811 [0.791, 0.827]	0.930 [0.919, 0.940]	0.853 [0.837, 0.866]	0.815 [0.800, 0.828]	0.804 [0.788, 0.819]
MMD	100	0.724 [0.696, 0.751]	0.791 [0.768, 0.812]	0.668 [0.644, 0.690]	0.729 [0.713, 0.745]	0.663 [0.643, 0.684]
MMD	10000	0.664 [0.650, 0.678]	0.740 [0.725, 0.755]	0.633 [0.617, 0.650]	0.670 [0.650, 0.689]	0.696 [0.678, 0.713]
IRM	1	0.918 [0.905, 0.930]	0.951 [0.940, 0.959]	0.878 [0.865, 0.889]	0.521 [0.498, 0.543]	0.464 [0.440, 0.486]
IRM	100	0.724 [0.713, 0.736]	0.752 [0.738, 0.765]	0.702 [0.690, 0.715]	0.496 [0.482, 0.509]	0.542 [0.528, 0.556]
GroupDRO	1	0.860 [0.843, 0.875]	0.941 [0.930, 0.949]	0.842 [0.828, 0.856]	0.646 [0.627, 0.664]	0.581 [0.558, 0.603]

Appendix B. Label-shift intervention: full results

[PENDING: Full R1 table: all shift levels × $\lambda_{\text{adv}} \in \{0.5, 2\}$ × five sets and five scores, with intervals; regression (a) details (shift coding, resampling unit, wild cluster bootstrap, VIFs), the cross-recipe comparison (b) and the leakage plots (c).]

Appendix C. BCN→HAM: label shift without DANN (post hoc)

ISIC 2019 contains two sites, Hospital Clínic Barcelona (BCN) and HAM10000 (HAM), both dermoscopic. Training on BCN with a lesion-disjoint held-out part of HAM as the OOD set, Mahalanobis AUROC is 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, below chance in all three seeds, while k-NN gives 0.658 $[0.627, 0.688]$. The reverse direction (HAM→BCN) gives 0.846.

Reweighting HAM to BCN's class mix recovers 92% of the distance to 0.5, no class-matched interval excludes 0.5, and dispersion ratios are close to one. The inversion is therefore a label-shift effect on the pooled score; we cannot exclude a small within-class contribution.

Appendix D. Domain-direction analysis

For each OOD image, the domain score is its projection on a linear ISIC-versus-`pad_adv` classifier fitted on the $\lambda_{\text{adv}} = 0$ model's features; the pull is the decrease in its Mahalanobis

Table 7: BCN→HAM at $\lambda_{\text{adv}} = 0$ (three seeds, lesion-clustered intervals). Class-matched: HAM images of class c against BCN test images of class c. Dispersion: median squared Mahalanobis distance to the image's own class centroid, HAM over BCN.

class	BCN test share	HAM share	class-matched Mahalanobis	class-matched k-NN	dispersion ratio
MEL	0.222	0.118	0.443 [0.362, 0.527]	0.620 [0.540, 0.693]	0.92
NV	0.306	0.667	0.483 [0.418, 0.546]	0.639 [0.584, 0.693]	0.97
BCC	0.215	0.052	0.498 [0.437, 0.561]	0.656 [0.585, 0.729]	0.99
AK	0.077	0.011	0.685 [0.570, 0.792]	0.595 [0.447, 0.733]	1.20
BKL	0.109	0.108	0.526 [0.445, 0.615]	0.623 [0.542, 0.705]	1.01
DF	0.021	0.012	0.577 [0.360, 0.768]	0.717 [0.593, 0.852]	1.05
SCC	0.042	0.020	0.407 [0.275, 0.528]	0.714 [0.594, 0.826]	0.90
pooled, unweighted			0.369 [0.337, 0.402]	0.658 [0.626, 0.688]	
pooled, HAM reweighted to BCN mix			0.489 [0.453, 0.524]		

distance to the nearest ID centroid, normalised by the median ID distance of each run, from $\lambda_{\text{adv}} = 0$ to $\lambda_{\text{adv}} = 2$ (matched seeds). The orthogonal control uses the magnitude of the deviation orthogonal to the domain direction. The placebo replaces the $\lambda_{\text{adv}} = 2$ model with a $\lambda_{\text{adv}} = 0$ model of a different seed.

Table 8: Spearman correlation between domain score and pull (latent z_ℓ^{norm}), and the excess over the seed placebo (paired bootstrap). Backbone: pre-projection features.

set	ρ_{dom}	ρ_{orth}	placebo ρ_{dom}	excess (latent)	excess (backbone)
`pad_heldout`	+0.344	−0.068	+0.288	+0.057 [+0.007, +0.105]	+0.331 [+0.283, +0.374]
Fitzpatrick17k	+0.419	−0.066	+0.205	+0.214 [+0.196, +0.232]	+0.383 [+0.364, +0.400]
MED-NODE	+0.433	−0.144	+0.106	+0.327 [+0.243, +0.407]	+0.349 [+0.256, +0.432]
SD-198	+0.467	−0.030	+0.203	+0.264 [+0.250, +0.278]	+0.449 [+0.436, +0.461]
SCIN	+0.333	+0.021	+0.150	+0.183 [+0.172, +0.193]	+0.239 [+0.227, +0.251]

Across λ_{adv} (cross-seed design), the correlation is already present at $\lambda_{\text{adv}} = 0.25$ and is slightly larger at $\lambda_{\text{adv}} = 2$ on four of five sets (differences 0.02–0.08). On the two BCN/HAM directions, the orthogonal deviation also predicts the pull, so the pattern is specific to the cross-modality setting.

Appendix E. Held-out domain probes

In the A4 recipe, the held-out linear and MLP probes still read the domain at $\lambda_{\text{adv}} \in \{0.5, 2\}$ (0.66–0.71) although detection inverts, so inversion does not require chance-level leakage. At $\lambda_{\text{adv}} = 0.25$ in the main recipe (observation after the fact, $n = 5$), the one seed whose held-out probe still reads the domain (0.67) does not invert on any set, three of the four seeds at chance invert on all four, and one inverts only on MED-NODE.

Appendix F. Importance-weighted alignment by prior estimate

EM estimation (Saerens et al., 2002) failed as implemented, including after a bug fix: in two of five seeds it placed mass on DF and VASC, which are absent from PAD, and its

Table 9: Domain probes (main recipe, best-validation checkpoint, five seeds). The first column includes PAD images the adversary trained on.

λ_{adv}	all of PAD (earlier work)	held-out linear	held-out MLP	held-out k-NN
0	0.927	0.827	0.921	0.935
0.25	0.614	0.538	0.547	0.586
0.5	0.581	0.506	0.500	0.548
1	0.563	0.504	0.500	0.560
2	0.555	0.502	0.500	0.578
4	0.636	0.516	0.500	0.570
8	0.663	0.550	0.523	0.608

Table 10: IWDAN at $\lambda_{\text{adv}} = 2$ (main recipe, last checkpoint, five seeds) by how the target class priors are obtained. TV: total variation from the true `pad_adv` priors. Leakage: unweighted linear probe on ISIC test against all of PAD; within-class: held-out within-class probe.

priors	TV	ID bal. acc	leakage	within-class	`pad`	Fitz	MED	SD
DANN (no weighting)	–	0.48	0.634	0.618	0.368	0.340	0.298	0.325
IWDAN (online)	0.43–0.55	0.48	0.642	0.596	0.513	0.508	0.350	0.488
IWDAN (50 labels)	0.10–0.14	0.39	0.806	0.583	0.619	0.741	0.606	0.734
IWDAN (true priors)	0	0.40	0.780	0.576	0.647	0.717	0.563	0.693

results are not used in any comparison. Black-box shift estimates computed on the final IWDAN checkpoints are a diagnostic, not a separate method (TV 0.65–0.76). [PENDING: IWCDAN, CDAN, $\lambda_{\text{adv}} = 0.5$ cells.]

Appendix G. Sign reversal and scoring-side variants

Reversing the Mahalanobis score with the direction chosen on `pad_heldout` at $\lambda_{\text{adv}} = 2$ gives 0.192 at $\lambda_{\text{adv}} = 0$ and 0.546–0.625 on `pad_heldout` and Fitzpatrick17k at $\lambda_{\text{adv}} \geq 0.5$; on MED-NODE it reaches 0.69–0.71 [PENDING: FPR@95 range]. Relative Mahalanobis at $\lambda_{\text{adv}} = 2$: Fitzpatrick17k 0.443, MED-NODE 0.376, SD-198 0.427, `pad_heldout` 0.439; per-class standardised Mahalanobis: 0.478, 0.340, 0.447, 0.424.

Appendix H. Clinical consequence

[PENDING: Per-class table on Fitzpatrick17k and SD-198 (MEL, BCC, SCC, AK): share accepted at the 95% ID threshold and share predicted NV with confidence above 0.9, at $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$, with n, intervals and the label map. `pad_heldout` has nine melanomas and is not used for this analysis.]

Appendix I. Negative results

Warning signals. Thresholds set at the 95th percentile of each statistic over $\lambda_{\text{adv}} = 0$ runs. A centroid-collapse statistic computed on in-distribution data only did not track

inversion (sensitivity 0.15–0.38; Spearman with AUROC −0.14 to 0.13). The share of unlabelled deployment images predicted as the majority class tracked it (Spearman −0.73 [−0.86, −0.51]; sensitivity 1.00, specificity 0.75 on twelve non-inverting runs with $\lambda_{\text{adv}} > 0$ under the Fitzpatrick17k label, 0.33 on six under the `pad_heldout` label). **Toy model.** Two-domain Gaussian mixtures with a majority class at 50% in the source and a label-shift level from 0 to 1, with a small encoder and gradient reversal, produced no inversion at any setting (Mahalanobis AUROC 0.57–0.66); leakage in the toy stayed near 0.5–0.58, so it may not reproduce the conditions of the real data.

Appendix J. Experiments that did not take place

Camelyon17 by hospital. In the domain-adversarial runs with hospital-disjoint held-out slides, the selected checkpoints were all from the warm-up epoch before gradient reversal took effect (15 of 15 runs), so these runs do not test the hypothesis. In runs whose selected checkpoint followed the warm-up (epochs 6–10), the hospital probe fell by at most 0.044 with in-distribution accuracy ≥ 0.99, and at $\lambda_{\text{adv}} = 300$ the probe fell by 0.081 while out-of-domain accuracy dropped from 0.91 to 0.70. In no setting did DANN remove hospital information while preserving the classifier. **Chest radiographs.** With NIH ChestX-ray14 and pediatric radiographs as the aligned domains, Mahalanobis AUROC on Shenzhen was 0.522 before any adaptation, below our headroom requirement, so we stopped before training adversarial models.

Appendix K. Image-level split (lesion leakage; for reference only)

The following results use the image-level ISIC split common in published work, in which 60% of test images share a lesion with training. They overstate in-distribution accuracy by about 0.20 and must not be compared with the main tables; we include them because earlier versions of this analysis used them.

Table 11: Image-level split (lesion leakage), best-validation checkpoint; leakage is the probe of earlier work. With the two non-adversarial controls. Five seeds at $\lambda_{\text{adv}} \in \{0, 0.25, 1, 2, 8\}$, three at $\lambda_{\text{adv}} \in \{0.5, 4\}$.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.571 ± 0.027	0.692	0.100	0.694	0.509 ± 0.023	0.460 ± 0.057	0.295 ± 0.028
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	5	0.591 ± 0.035	0.695	0.097	0.735	0.457 ± 0.026	0.414 ± 0.086	0.258 ± 0.012
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391